Session-III: Participatory Hydrological Monitoring (PHM): August 1st week

Program Schedule

S.No	Time	Content	Method	Materials	Expected outcomes
1	10 min.	Recap.	Large group discussion	Charts, markers and clips.	Participants recollect the contents of second session.
2	20 min.	Relevance of hydrological data.	Large group discussion and pictures.	Pictures, charts and markers.	Participants identify the need to collect PHM data.
3	15 min.	Measuring devices.	Large group discussion and demonstration	Water level indicator, 100 litres drum, and stop watch.	Participants understand the application and usage of various measuring devices.
4	75 min.	Data collection procedures and dissemination.	Demonstration and large group discussion	Charts, markers, pens.	Participants measure the water levels, discharge of borewell and rainfall.
5	15 min.	Group dynamics – Book-keeping.	Game	Twenty different kinds of items.	Participants will discuss the importance of recording data in HMR book.
6	10 min.	HESA – Water Levels.	Large group discussion	Charts and markers.	Participants analyze water levels.
7	20 min.	Short Study: Inflow and outflow of the HU.	Demonstration	Soil, heap of mud.	Participants observe the inflow and outflow of rainwater and discuss the rate of infiltration into ground and the amount of out flow.
8	10 min.	Homework: PHM data measurement.			Farmers observe water levels in their respective villages.
9	10 min.	Evaluation.	Large group discussion	Charts and markers.	Participants share their opinions on the conduct of FWS sessions and methods used in the session.

Learning objectives

By the end of the session, participants will be able to:

- Recap the previous session,
- Discuss hydrological data and its relevance,
- Understand the usage of measuring devices,
- Discuss the procedures of data collection and dissemination,
- Understand the importance of book-keeping,
- Analyze the relationship between water levels and rainfall (HESA), and
- Discuss the need to observe water levels.

Process

1. Recap Time: 10 min.

Objective : Participants recap the previous session.

Content : Recap session two contents.

- HU walk (HESA): Field walk in Hydrological Unit area and water cycle.
- Water cycle of the Hydrological Unit.
- Importance of leadership qualities.

Method : Large group discussion

Materials : Charts and markers.

Process

- Welcome the participants to third session.
- In the large group, encourage each participant to share a topic that was discussed in previous session and the learning from that one.
- As you facilitate the recap, write participant's responses on the chart paper.
- Briefly discuss the contents and the learning from the previous session.

2. Relevance of hydrological data Time: 20 min.

Objective : Participants discuss the relevance of hydrological data.

Content :

- Timeline of depleting water levels and adaptation of tools.
- Relevance of PHM activities to water levels and rainfall.

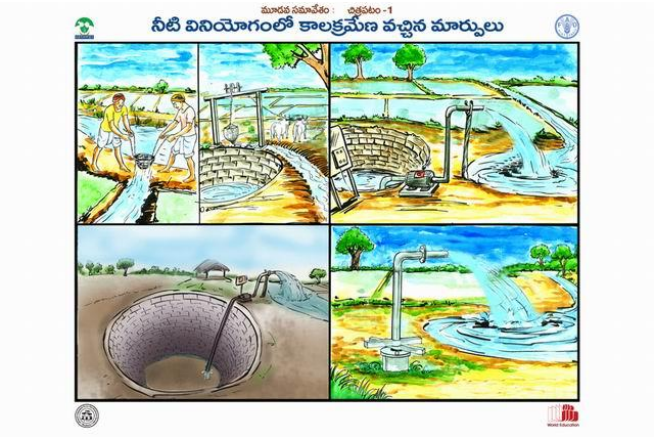
Method : Large group discussion

Materials : Picture, charts, and markers.

Process

Water level and discharge measurement

- Start the discussion stating that: "We are dependent on agriculture. Could you please recollect, what were the sources of irrigation 50 years ago?".
- Write responses of the participants on a chart paper..
- Show the picture depicting timeline of depleting water levels and irrigation practices.



- Continue the discussion stating that: "We used to raise crops based on water availability in open wells and tanks. Now, we are depending on borewells for irrigation and drinking purposes. This shift is mainly due to drying of surface water bodies, lack of adequate rainfall, and deforestation".
- Pose a question in large group if farmers are happy with the present state of agriculture using borewell irrigation and what are the challenges in it?
- List the challenges on a chart paper.
- Encourage a discussion in large group by asking the following questions:
  - "Do you think water levels remain static in your borewell?
  - How do you know at what level water is available in your borewell?
- List the participant's responses on a chart paper and summarize by sharing that it is good to measure water levels and discharge (Time required to fill 100 litres drum) of the borewell to understand water levels in a borewell.

Rainfall measurement

- Start the discussion with the following words:
- All of us are aware that the primary source for irrigation in agriculture is rainfall. We will have enough water available in tanks and borewells for irrigation when there is adequate rainfall.
- Pose the following question in large group to extract the indigenous knowledge of farmers: "How do you estimate rainfall as adequate, moderate or poor?"
- Write participants responses on a chart paper and continue the discussion stating that these assumptions cannot give the exact picture.
- Summarize the discussion by stating that it is always useful to be aware of a scientific method to measure exact rainfall received and to analyze whether the received rainfall is adequate, moderate or poor.
- Conclude the discussion by sharing that measuring water levels, rainfall will enable us to understand the groundwater dynamics and the relation between rainfall, water levels and discharge of the borewell.

Tips to facilitator

Local farmer's assumptions

- Farmers in the village assume that presence of earthworms in their field is an indication of adequate rainfall.
- Sprouting of mushrooms in their field's vicinity is taken as another indication.
- Though these are indicative of presence of moisture in soil, it is not possible to measure the exact quantity of rainfall received by these methods.

3. Measuring devices Time: 15 min.

Objectives : Participants discuss the usage of measuring devices.

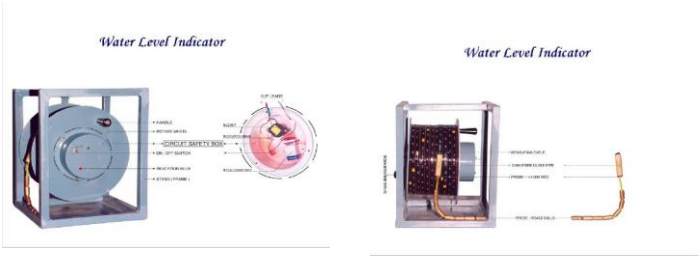
Content : Measuring devices.

Method : Large group discussion and demonstration

Materials : Water level indicator, drum, stop watch, collecting jar, measuring jar.

Process

- Start the discussion stating that: "We have just finished discussing the need to measure water levels and rainfall. Let us discuss about the methods or tools used to measure rainfall and water levels of borewells".
- Ask the following questions in large group:
  - How can we know at what level, water is in your borewell? And what are the usual methods you use to know the water levels?
- Write participant's responses on a chart and introduce Water Level Indicator (WLI).
- Demonstrate the functioning and different parts of water level indicator.
  - The WLI has a cable and numbers written on it in 'feet' or 'meters.'



  - When both ends of the probe of WLI touches water surface, it gives a 'beep' sound. At this point, we need to observe the reading on the cable and note the same in HMR book.
  - Before taking the reading, ensure the exact contact point of probe with water surface.
  - Next, introduce 100 lt. drum and stop watch used to measure the time taken for bore well to fill 100 lt. drum (Called 'discharge').



- Next, encourage rain gauge volunteer to share the instruments used to measure the rainfall received. Show the rain gauge, collecting jar and measuring jar. Share that the measuring jar will be calibrated into millimetres.



4. Data collection procedures and dissemination Time: 75 min.

Objective : Participants demonstrate the process of measuring water levels, discharge of the borewell and the rainfall.

Content :

- Procedures of measuring water levels, discharge and rainfall.

Method : Large group discussion and demonstration

Materials : Water level indicator, drum, stop watch, collecting jar, measuring jar.

Process

Water level measurement

- Take the participants to a nearby borewell.
- Encourage rain gauge volunteers and observation borewell volunteers to explain the process of measuring rainfall and water levels respectively.
- Explain that it is an observation borewell fitted with one inch HDPE pipe to insert WLI.



- Demonstrate how to insert the WLI and ask the participants to notice the sound when both the ends of the probe touch the water surface.
- Demonstrate how to measure the water levels.
- Discuss the time interval between Static Water Level (SWL) and Pumping Water Level (PWL). It should be minimum three hours after switching on the pump or motor.
- Ask participants in large group the reason for measuring PWL after a lag of three hours?
- Summarize the discussion by stating that water in the borewell is agitated during the pumping process. If a measurement is taken immediately after switching on the

borewell, the result will be false due to accumulated water. It will take a minimum of three hours for it to settle down to a stable level. So, it is advisable to measure after three hours of pumping.

- Encourage the participants to practice measuring water levels.

Discharge measurement

- Next, ask the participants if it is possible to measure the quantity of water drawn from a borewell.
- Show the 100 litres calibrated drum and stop watch and demonstrate its usage and process of measuring the quantity of water.
- Explain that if we know the time taken to fill a 100 litres drum, we can estimate the quantity of water pumped out per hour and per day.
- Encourage the participants to practice measuring drum discharge.



Rainfall measurement

- Next, take the participants to the nearby rain gauge station and explain the process of measuring rainfall and time for measuring rainfall.
- Demonstrate the measuring of rainfall.
- Encourage the participants to practice the measuring rainfall.
- Next, Select four volunteers and explain the same.



Data dissemination – Types of data display boards

- Next, initiate a discussion on the importance of data dissemination to the larger audience in the village by posing the following questions:
  - Is the data that has been generated useful to our co-farmers (borewell users)?
- If it is useful, how can you effectively disseminate this information to our co-farmers? Write the participant's responses on chart paper.

Tips to facilitator
Encourage participant's to explore different ways like: • Sharing in women groups by women members of the GMC or FWS participants. • Discuss and share in regular GMC meetings. • Sharing in regular meetings of village panchayat. • Discussion in common gatherings at village level. • Use of posters, wall paintings.

- Introduce different kinds of dissemination boards and explain the process of recording data on them.
- Emphasize the importance of the location for displaying dissemination boards. This location should be ideally visible to everyone in the village.



Tips to the facilitator
Types of data dissemination boards: • Water level type. • Rain fall type. • HU type.

5. Group dynamic: Book-keeping Time: 15 min.

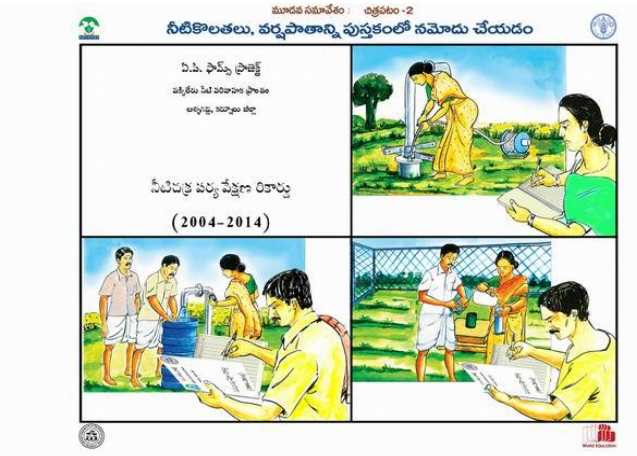
Objective : Participants will discuss the importance of recording data in HMR book.

Materials : Twenty different types of items (Ex: pen, book, lid, paper, seed, stone, dried leaf etc).

Method : Game

Process

- Place the 20 different items on the floor and cover it.
- Ask the participants to watch the 20 items for two minutes and cover the items.
- Ask the participants to write the items individually on a piece of paper.
- Participants may not list all the items.
- Ask the following question to generate the discussion. "If it is difficult for us to recall the items that we have just watched, will it be possible to for us to recollect the water levels data which was collected 15 days ago?"
- Conclude by saying that it is important to maintain the record of water levels and rainfall data. It is critical for the conduct of crop water budgeting exercise and for making collective decisions around crop-plans.



- After completion of discussion on the importance of data recording in Hydrological Monitoring Record (HMR) books, introduce the HMR book and the contents in it to the participants.
- Distribute the photocopies of the HMR book and encourage the participants to practice recording of data.
- Discuss the content to be added in mentioned rows and columns of the HMR book and request the participants to record the collected data in appropriate columns.
- Initiate a discussion in large group on the frequency of measuring water levels, discharge and rainfall. Write participants responses on a chart paper and conclude by sharing the following information with the participants.
- Measure the rainfall early in the morning before 8 am whenever we receive rainfall. This is because the rain collected in the rain gauge might start evaporating and give rise to false results.
- Measure the water levels and discharge every fort night and record in HMR book for future reference.

6. HESA-Water levels Time: 10 min.

Objective : Participants analyze water levels.

Materials : Charts and markers.

Method : Large group discussion

Process

- Ask the participants in large group if they have been measuring water levels without any difficulty?
- Clarify their doubts (if any).
- Pose a question in large group: 'Did you observe any change in water levels or discharge of the borewell?
- Summarize the discussion by analyzing the reasons for these fluctuations. Use HESA frame work for analysis.

7. Short Study - Inflow and outflow of the HU Time: 20 min.

Objective : Participants will observe the inflow and outflow of rainwater and discuss the rate of infiltration into ground and the amount of out flow.

Rationale : Usually, farmers believe that a large portion of rain water infiltrates into the ground. In reality, only a small fraction percolates into ground. Most of the rainwater flows into oceans through rivers. The following experiment will help demonstrate:

- How rain water flows into a Hydrological Unit?
- How only a small percentage of it infiltrates into the ground?
- How a major percentage of water flows out of Hydrological Unit?

Here are two alternate ways of demonstrating or generating discussion on this.

Model One : Use of soil

Materials : 45 x 30 cm tray, polythene cover, empty bottle, 4 liters of water, soil, small stones, small branches of tree, and pins.

Method :

- Wet the soil with water.
- Prepare a model of the Hydrological Unit in the tray using wet soil. Use the stones to represent hills. Place leaves and small branches in the mud to represent forest area in the Hydrological Unit area.
- Put a hole to the tray towards slope portion

Usage :

- Pour 2 liters of water in polythene cover and hold it above the tray.
- Use pins to pierce the polythene cover to simulate rain.
- The water that falls on the tray comes out of it through the slope.

Note**:** Wet the soil with more water to demonstrate more out flow.

Discussion Questions:

- How does rain water flow in a Hydrological Unit?
- What percentage of it infiltrates into the ground?

Model two *:* Use a heap of mud

Materials **:** Plastic tray, polythene cover, vessel (to receive water), pins, mud, small stones and leaves.

Method :

- Fill the plastic tray up to the brim with wet soil.
- Arrange the soil to form slope. Use the stones to represent hills.
- Place leaves and small branches in the mud to represent forest area in the Hydrological Unit area.
- Show the origin of the stream at height, and the end point of the stream towards the slope of the tray.

Usage

- Keep the model at a height so that all the farmers can see.
- Hold the polythene cover filled with water above the tray. Use pin head to pierce the polythene cover to simulate rain.
- Rain Water (in-flow) covers most part of the area in the model, flows and comes out of the tray (out-flow).
- Explain to them that rain water falling on hills and upper reaches of the area will enter rivers via rivulets and streams and finally will enter into the seas (confluence).

Discussion Questions:

- How does rain water flow in a Hydrological Unit?
- What percentage of it infiltrates into the ground?

8. Homework – PHM data measurement Time: 10 min.

- Encourage the participants to observe the water levels in their respective villages and request them to come back to the next session with their questions and observations.

9. Evaluation Time: 10 min.

Pose the following questions to evaluate the day's learning:

- Is there a need to share the importance of measuring water levels, discharge and rainfall in GMC meetings? Why?
- Please list the measuring devices to measure water levels, discharge and rainfall.
- Please list the different types of dissemination boards and their usefulness.